EE352 Automatic Control

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Deep Vision, Bright Future

Course aim

- To provide understanding of analysis and modelling of continuous time control systems and controller design techniques

Intended Learning Outcomes (ILO)

At the completion of this module, the students should be able to

1. define, identify signal flows of control systems and analyze the impact of feedback on the performance.
2. differentiate Single-Input-Single-Output (SISO) Linear Time Invariant (LTI) control systems and model them using linear differential equations and analyze stability.
3. analyze control systems using time domain responses for standard test signals and frequency domain tools, namely Bode diagrams, Nayquist diagrams and Root Locus diagrams.
4. design controllers to meet time and frequency domain specifications using P, PI, PD, PID, Pole placement, Pole-Zero cancellation, Phase Lead, Phase Lag and Phase Lag-Lead compensation methods and interpret their merits and demerits.
5. analyse the impact of controllers on the performance, implement the controller designs using analogue electronics and solve practical issues using standard techniques.

Course contents

- Control system fundamentals
- System modeling
- Modeling of practical systems
- Time domain analysis
- Frequency domain analysis
- Controller design in continuous domain

Assessment

Assessment	Percentage	Marks
Continuous Assessment	20	
Assignments		5
Laboratory work		15
Written Examination	80	
Mid-Semester		30
End-Semester		50

Course Participation Definition

- Online sessions held = A
- Activities submitted by the student = a
- Total laboratories held = B
- Laboratories attended by the student = b
- Total Quizzes & Assignments held = C
- Quizzes & Assignments attended by the student = c

- Course Participation = $0.4\left(\frac{a}{A}\right) + 0.5\left(\frac{b}{B}\right) + 0.1\left(\frac{c}{C}\right) \geq 0.8$

List of experiments

1. Computer Aided Control Systems Analysis Tools
2. Classical Controller Design Approaches
3. Modern Controller Design Approaches

List of Assignments

1. Time domain analysis of Control Systems and Design
2. Frequency domain analysis of Control Systems and Design

Quizzes

- There will be 2 to 3 quizzes spread over the semester.

References

1. B.C. Kuo, “Automatic Control Systems”, Seventh edition, Prentice-Hall International, Inc., Englewood Cliffs, N.J., 1995.

2. G.F. Franklin, J.D. Powell, M.L. Workman, “Digital Control of Dynamical Systems”, Second Edition, Addison-Wesley Publishing Company, Inc., 1990.